

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.3419603906715553, median 0.31822184301485784, std: 0.19167248449897503
Reprojection error (cam1): mean 0.3884373233760441, median 0.36646164293582734, std: 0.20543268525643857
Gyroscope error (imu0): mean 0.2898519508358399, median 0.22641521290431313, std: 0.22680834228518373
Accelerometer error (imu0): mean 0.18249026834192528, median 0.13949231337875379, std: 0.1527423419580261

Residuals

Reprojection error (cam0) [px]: mean 0.3419603906715553, median 0.31822184301485784, std: 0.19167248449897503
Reprojection error (cam1) [px]: mean 0.3884373233760441, median 0.36646164293582734, std: 0.20543268525643857
Gyroscope error (imu0) [rad/s]: mean 0.014492597541791994, median 0.011320760645215658, std: 0.011340417114259186
Accelerometer error (imu0) [m/s^2]: mean 0.03649805366838505, median 0.027898462675750753, std: 0.030548468391605223

Transformation (cam0):

T_ci: (imu0 to cam0):
[[-0.01186709 -0.99789644 0.06373281 0.02763896]
[-0.99991853 0.01154317 -0.00544835 -0.00932579]
[0.00470121 -0.06379227 -0.99795213 -0.05456692]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):
[[-0.01186709 -0.99991853 0.00470121 -0.0087405]
[-0.99789644 0.01154317 -0.06379227 0.02420752]
[0.06373281 -0.00544835 -0.99795213 -0.05626749]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.014584348645694935

Transformation (cam1):

T_ci: (imu0 to cam1):
[[-0.01159116 -0.99746857 0.07015767 -0.02933897]
[-0.99991936 0.01119838 -0.00598934 -0.00867954]
[0.00518853 -0.07022144 -0.99751793 -0.05461192]
[0. 0. 0. 1.]]

T_ic: (cam1 to imu0):
[[-0.01159116 -0.99991936 0.00518853 -0.00873556]
[-0.99746857 0.01119838 -0.07022144 -0.03300243]
[0.07015767 -0.00598934 -0.99751793 -0.05247]
[0. 0. 0. 1.]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
0.014990324045671789

Baselines:

Baseline (cam0 to cam1):
[[0.99997923 -0.00030597 -0.0064377 -0.05733149]
[0.00030959 0.99999979 0.00056187 0.00066834]
[0.00643753 -0.00056385 0.99997912 -0.00022932]
[0. 0. 0. 1.]]

baseline norm: 0.057335840209653655 [m]

Gravity vector in target coords: [m/s^2]
[-0.26733778 -9.80232708 -0.10647653]

Calibration configuration

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cam0

Camera model: pinhole
Focal length: [424.53652127374767, 424.4843320181217]
Principal point: [320.56931003970544, 260.8367083131217]
Distortion model: radtan
Distortion coefficients: [-0.36837350252522294, 0.10962688072302687, 0.0006862395588788541,
-0.001141627796693045]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.04 [m]
Spacing 0.012 [m]

cam1

Camera model: pinhole
Focal length: [426.5373643938949, 426.5188303796709]
Principal point: [320.5058702981205, 255.76821443915068]
Distortion model: radtan
Distortion coefficients: [-0.36763969027571325, 0.10729935167201371, 0.0011742047923574235,
-0.0001497684283195815]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.04 [m]
Spacing 0.012 [m]

IMU configuration

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IMU0:

Model: calibrated
Update rate: 100.0

Accelerometer:

Noise density: 0.02

Noise density (discrete): 0.19999999999999998

Random walk: 0.003

Gyroscope:

Noise density: 0.005

Noise density (discrete): 0.049999999999999996

Random walk: 2e-05

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

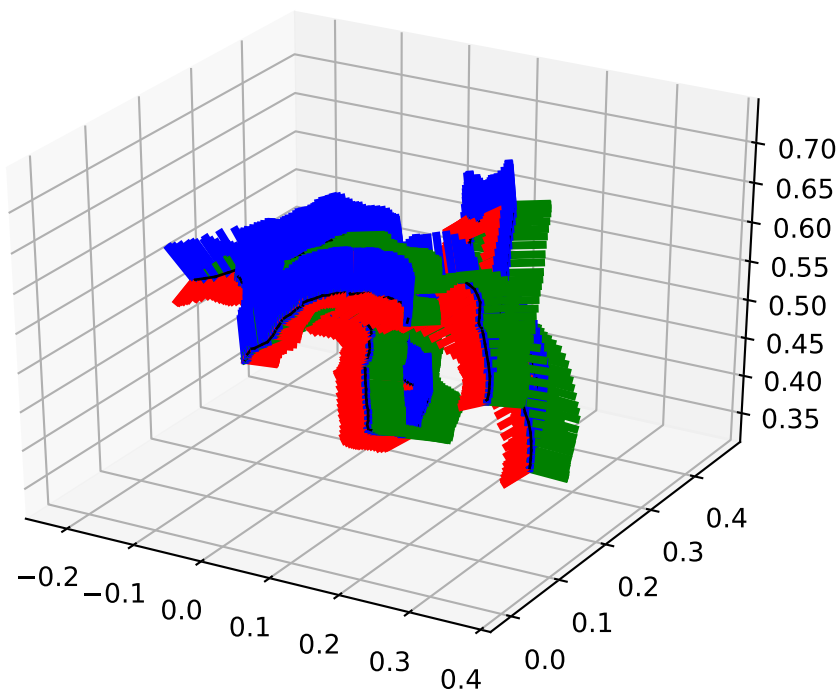
[0. 1. 0. 0.]

[0. 0. 1. 0.]

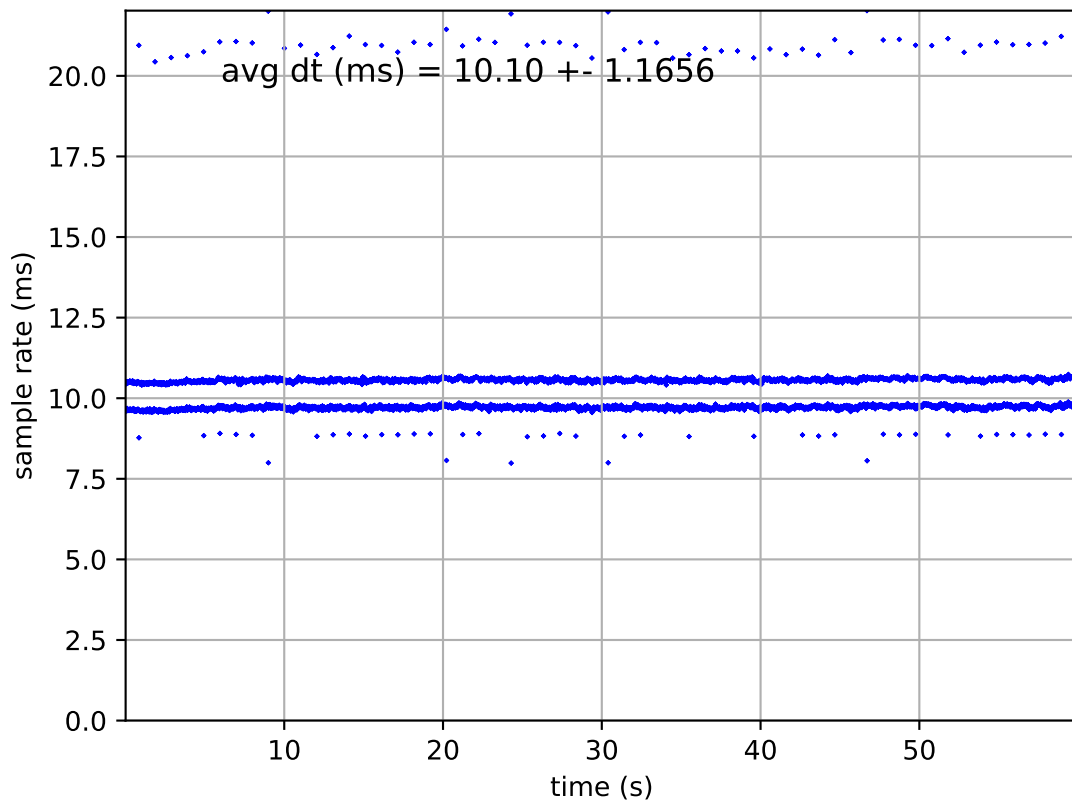
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

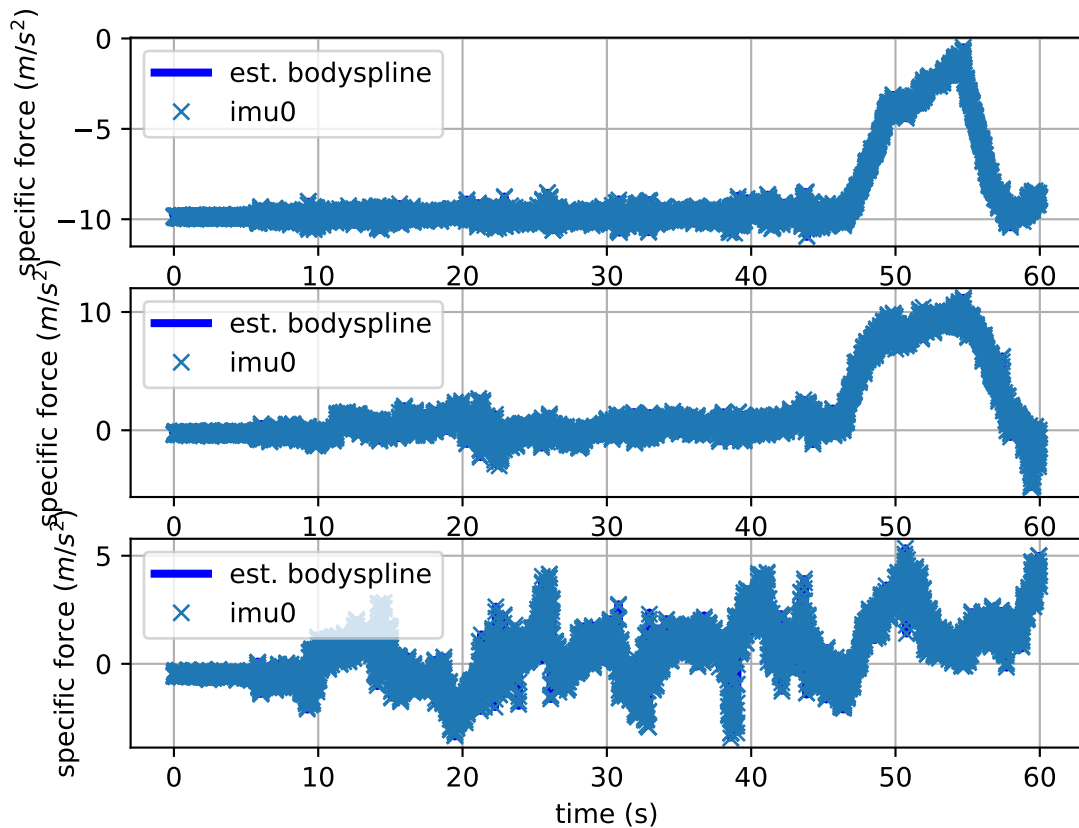
imu0: estimated poses



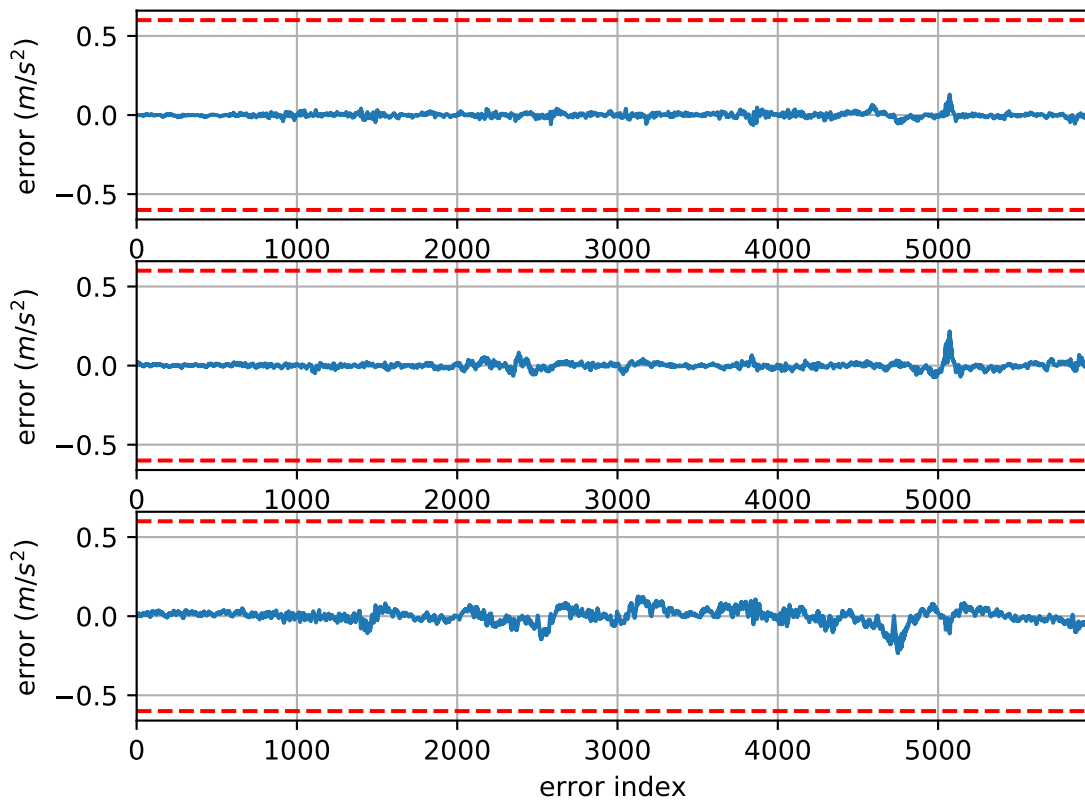
imu0: sample inertial rate



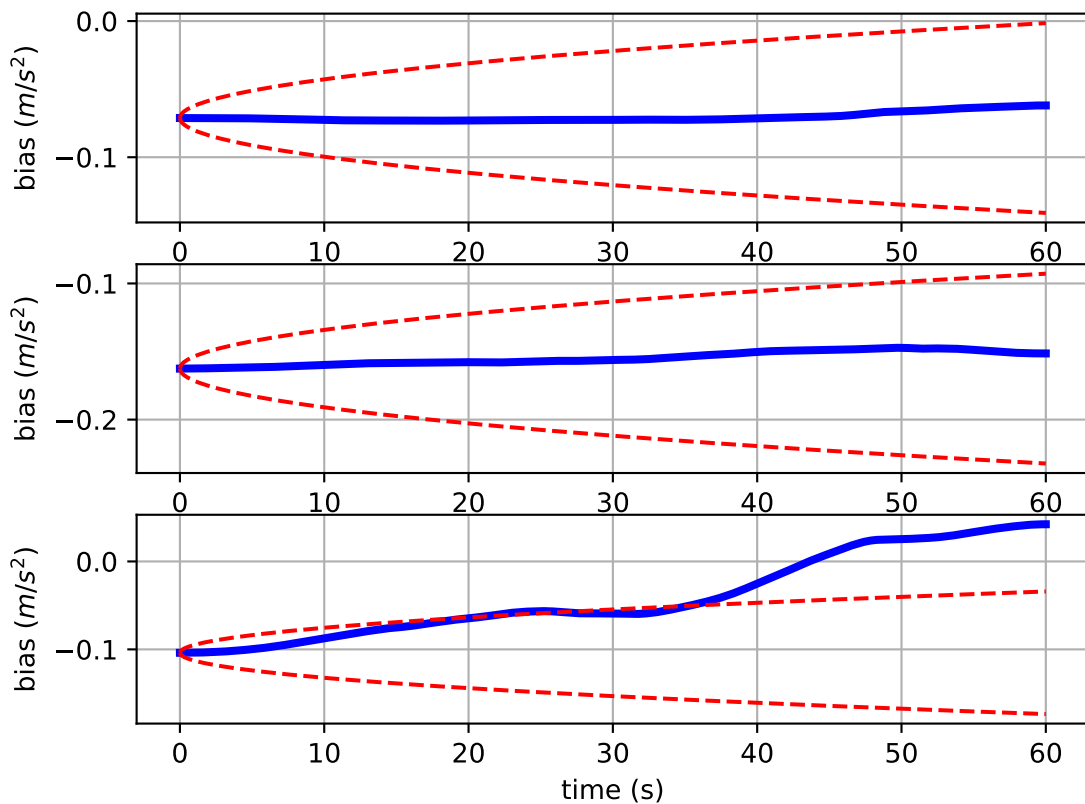
Comparison of predicted and measured specific force (imu0 frame)



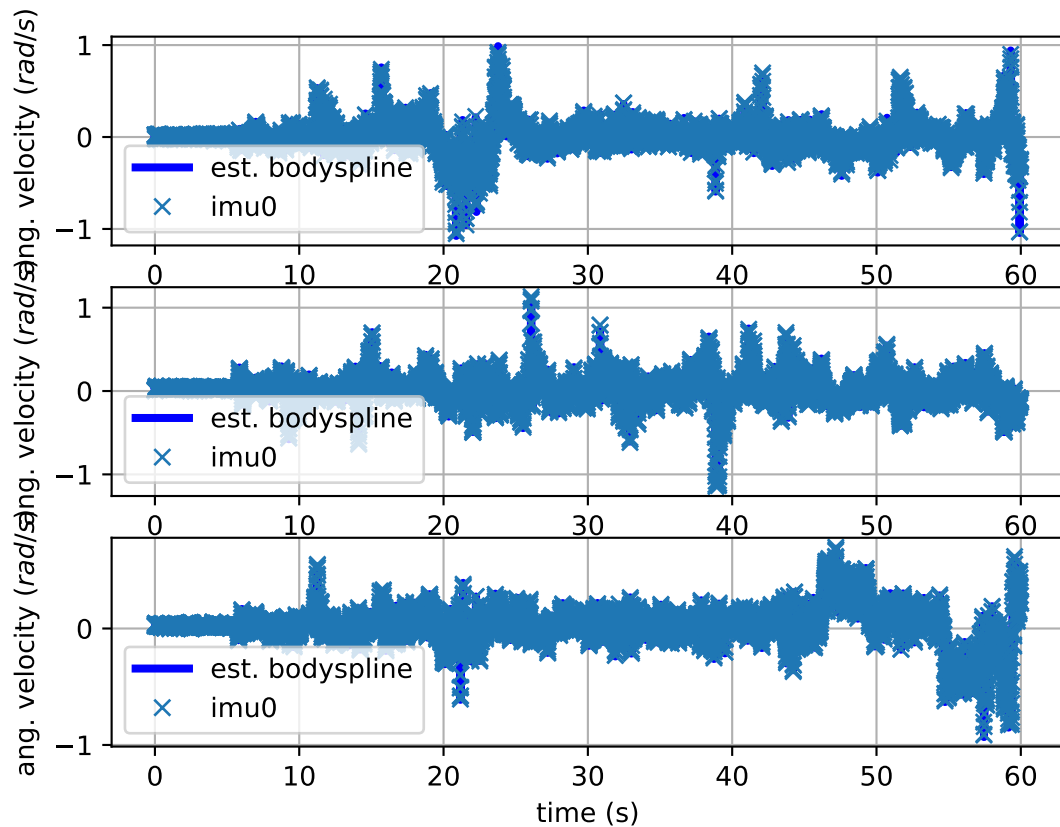
imu0: acceleration error



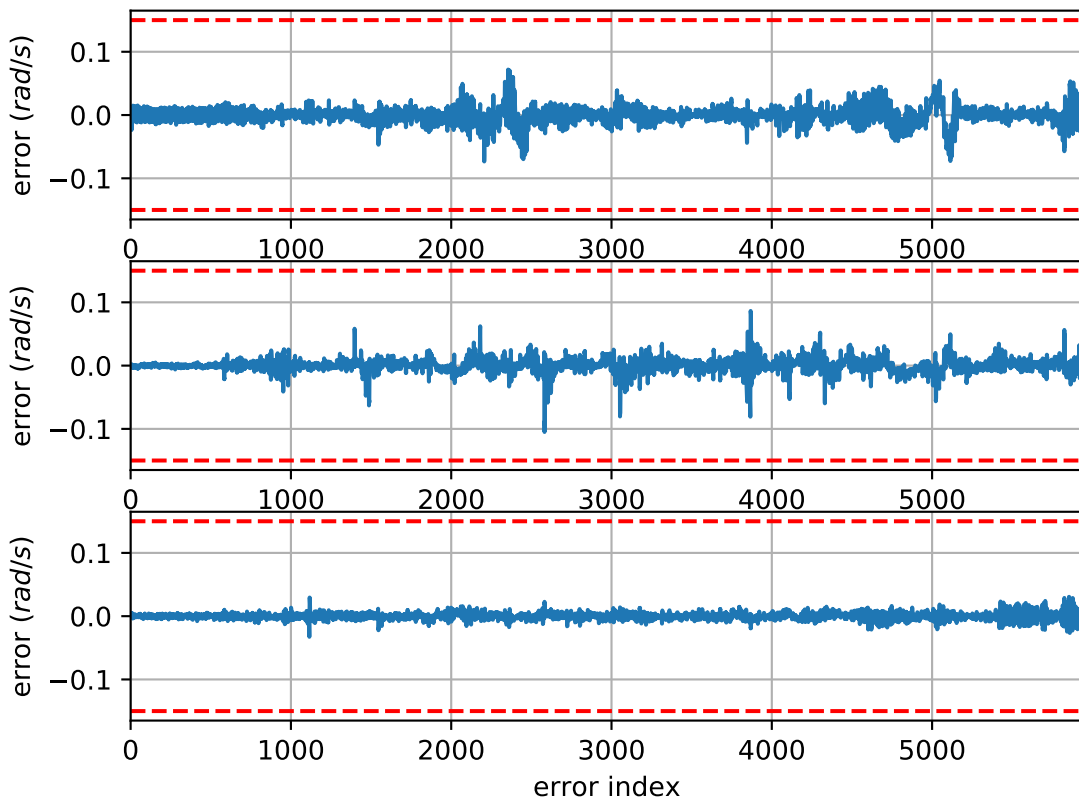
imu0: estimated accelerometer bias (imu frame)



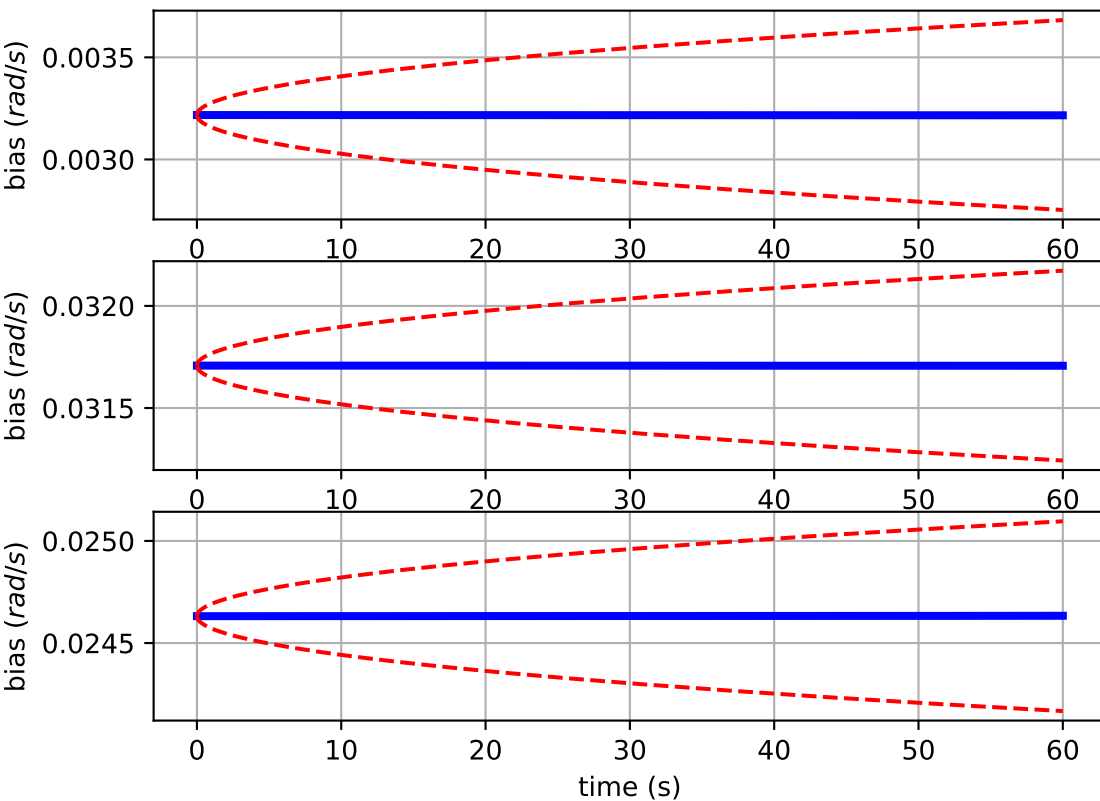
Comparison of predicted and measured angular velocities (body frame)



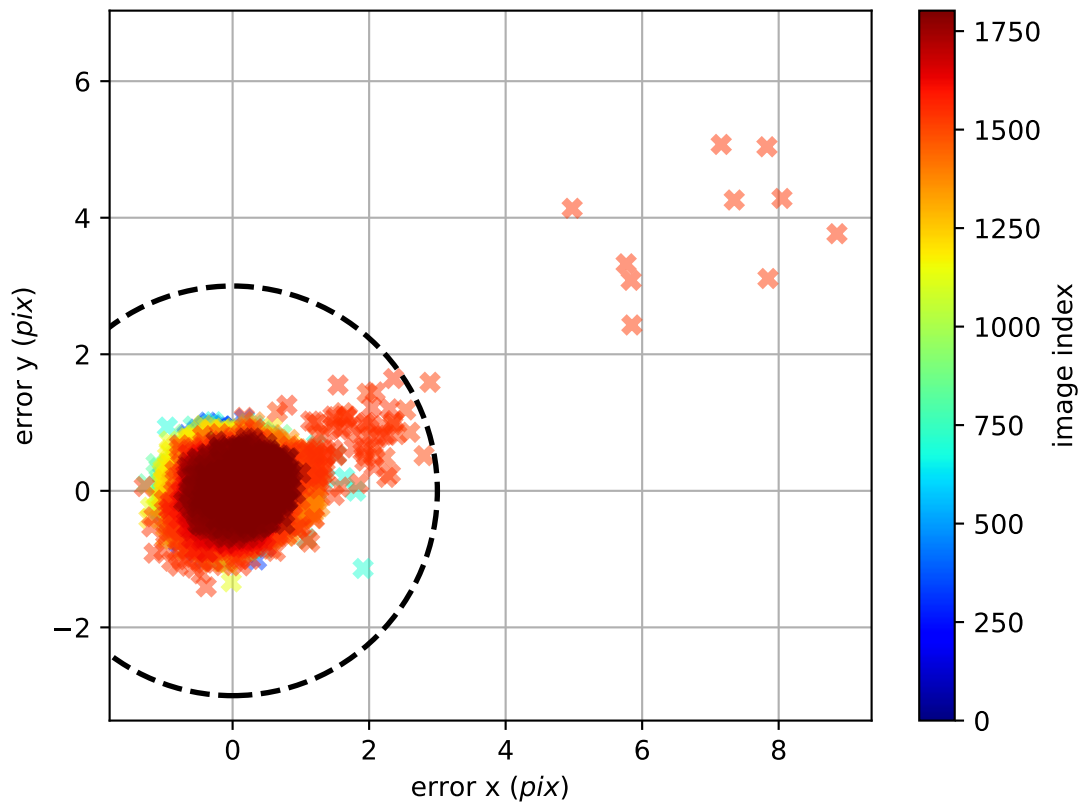
imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors



cam1: reprojection errors

